

ORDI: Orbit-Aware Redundant Distributed Inference for LEO Earth Observation Constellations

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Abstract—Low-Earth-orbit Earth-observation constellations collect more imagery than intermittent ground contacts can promptly downlink, motivating distributed onboard inference for time-critical alerts. Yet changing contact geometry, thermal and battery throttling, and correlated orbital-plane failures make static task placement unreliable. We present ORDI, an orbit-aware rolling-horizon scheduler that jointly considers predicted contacts and each satellite’s compute, energy, temperature, queue, and availability state. ORDI ranks image tiles by urgency-weighted utility, selects feasible primary helper-aggregator pairs, and adds fault-disjoint backups only when their marginal reliability gain exceeds energy, communication, and replication costs. Detected failures, missed contacts, and stragglers trigger local replanning. We compare ORDI with eight baselines across fault type and intensity, deadline, constellation scale, correlated outage, and backup-cap studies, using a uniform conservative 25° ground-link elevation mask. Under nominal stochastic loss, ORDI achieves the highest utility while using 40% fewer replicas and 40% less energy than full replication. Under one-plane outages it remains within 3.5 percentage points of full replication’s realized miss ratio with 44% less energy. Its greedy solution attains slightly higher utility than an ILP reference with $5.8\times$ less inter-satellite traffic. Tests on real-world NASA FIRMS wildfire requests further preserve the synthetic study’s utility ordering, supporting selective redundancy under operational demand conditions.

Keywords—LEO constellations, Earth observation, distributed inference, fault-tolerant scheduling

II. Background and Related Work

A. Onboard Inference and Orbital Edge Computing

EO satellites are bandwidth-starved: ground contact windows last only 5–10 minutes per pass, and many operators schedule only high-latitude stations, so a satellite over the southern hemisphere may wait most of an orbit to downlink. Onboard inference attacks this at the source, reducing a megabyte-scale tile to a kilobyte-scale result and shifting the bottleneck from downlink to onboard compute, which is in turn bounded by a small spacecraft’s power and thermal envelope. That this is practical is now established by flight programs across agencies and industry: ESA’s Φ -Sat-1 flew the first on-board deep-neural-network demonstrator [1], re-trainable payloads have been uplinked and run in orbit on a 1 W VPU [2], the NASA/HPE Spaceborne Computer-2 ran AI workloads on the ISS with software radiation hardening [3], and the Tiansuan constellation reported systematic in-orbit COTS measurements [4] showing accelerators throttled well below nameplate throughput by thermal and battery

limits. Earlier work established the paradigm (orbital edge computing [5], onboard model specialization [6]), and recent surveys catalogue it [7]. This body of work is almost entirely single-satellite inference; ORDI addresses the next question of distributing one task across satellites when a single satellite’s compute or downlink cannot meet the deadline.

B. Multi-Satellite Offloading and Placement

As LEO mega-constellations mature, several systems schedule computation across satellites. Serval bifurcates near-real-time EO queries on one satellite’s resources [8]; SECO optimizes latency or energy for multi-satellite placement [9]; and LEOEdge co-designs satellite-ground AI-inference placement with per-satellite adaptive models [10]. These approaches optimize placement under nominal availability but assume scheduled resources survive; we quantify the resulting reliability loss and show that structured redundancy recovers utility efficiently.

C. Priced Redundancy under Correlated Failures

Tiling makes inference embarrassingly parallel: a 4×4 grid yields 16 independent units that ISLs can spread across a neighborhood. But every replica’s success requires the source to emit the tile, each ISL hop to hold, the helper to survive compute, and the aggregator’s downlink to materialize—and these events are correlated, since satellites in one plane share geometry, eclipse cycles, and deployment batch, so a plane-wide outage disables many helpers at once. Tail-tolerant duplication [11] and coded computation [12], [13] address this in datacenters, but assume cheap bandwidth and independent failures. In orbit, full replication costs roughly double the energy and ISL traffic that a 20–30 Wh-class spacecraft cannot afford, and correlation breaks the coding assumption. The design space between zero redundancy and full replication, priced by orbital state and placed against the actual failure correlation, is the territory ORDI explores.

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